FED4FED: Statistical Fairness Evaluation Report

Privacy-Preserving Verification for Federated Learning Models

FAIRNESS VERDICT: FAIR (Performance Homogeneous)

Under the Conservative Joint Policy (Disparity detected if either Alg 1 or Alg 2 rejects H0), the global federated model demonstrates statistically fair performance. Neither test found significant evidence to reject the null hypothesis of equal performance at alpha = 0.05. Alg 1 p-value: 0.3306, Alg 2 p-value: 0.3374.

1. Experiment & Architecture Configuration

Experiment ID:	sst_fair_20261020_224957	**Date / Time:**	2026-10-20 23:03:18
Dataset:	SST	**Scenario:**	Fair
Clients (K):	10	**Seed:**	42
FL Rounds:	10	**Final Val Accuracy:**	0.8281
Partition Method:	uniform	**Bootstrap Replications (B):**	300

2. Statistical Fairness Hypothesis Tests

Test Algorithm	Statistic	df	p-value	Significance (α)	Decision
Algorithm 1 (Graybill-Deal / T1)	10.2504	9	0.3306	0.050	**Fail to Reject (Fair)**
Algorithm 2 (Welch-Type F)	1.1315	(9, 810.5)	0.3374	0.050	**Fail to Reject (Fair)**

3. Algorithm 3: Global Performance Confidence Interval

Status: Computed under homogeneous fairness condition.
Confidence Level: 95.0%
Pooled Performance Estimate (Graybill-Deal): 0.8313
Confidence Interval: [0.8156, 0.8469]
Standard Error: 0.0080 (Effective df = 810.5)

4. Sample Decentralized Client Performance Statistics

Client ID	Test Samples (n_i)	Observed Metric	Bootstrap Mean	Variance (S_i^2)	Std Error (SE)
Client 00	221	0.8597	0.8603	0.000518	0.0228
Client 01	221	0.8190	0.8175	0.000617	0.0248

Client 02	221	0.8643	0.8618	0.000532	0.0231
Client 03	221	0.8507	0.8527	0.000503	0.0224
Client 04	221	0.8190	0.8188	0.000712	0.0267
Client 05	221	0.8371	0.8385	0.000645	0.0254
Client 06	221	0.8054	0.8063	0.000689	0.0262
Client 07	221	0.8190	0.8191	0.000689	0.0263

Scientific Methodology: Fed4Fed privacy-preserving statistical homogeneity testing using non-parametric bootstrapping, Graybill-Deal common mean estimation, and Welch-type heteroscedastic F-testing. Generated by FED4FED Research Suite v1.0.0.